

Sahrdaya College of Engineering and Technology, Kodakara
Second Semester B.Tech Degree (R/S) Examination June 2023

MAT102 (2019 Scheme)

Vector Calculus, Differential Equations and Transforms

PART A

- 1) Find the parametric equation of the tangent vector of the curve $\vec{r}(t) = t^2 \hat{i} + 2t^3 \hat{j} + 3t \hat{k}$ at $t=1$.

$$\vec{r}'(t) = 2t \hat{i} + 6t^2 \hat{j} + 3 \hat{k}$$

$$\vec{r}'(1) = 2 \hat{i} + 6 \hat{j} + 3 \hat{k}$$

$$\underline{\underline{x(t) = 2, \quad y(t) = 6, \quad z(t) = 3}}$$

- 2) Find the directional derivative of $f(x,y) = x e^y$ at $(1,1)$ in the direction of the vector $\hat{i} - \hat{j}$.

$$f_x = e^y, \quad f_x(1,1) = e$$

$$f_y = x e^y, \quad f_y(1,1) = e$$

$$\hat{u} = \frac{\hat{i} - \hat{j}}{\sqrt{1^2 + (-1)^2}} = \frac{\hat{i} - \hat{j}}{\sqrt{2}}, \quad u_1 = \frac{1}{\sqrt{2}}, \quad u_2 = \frac{-1}{\sqrt{2}}$$

$$\text{Directional derivative} = f_x u_1 + f_y u_2$$

$$= e \cdot \frac{1}{\sqrt{2}} + e \cdot \frac{-1}{\sqrt{2}} = \underline{\underline{0}}$$

- 3) Use Green's theorem to evaluate $\oint_C x dy - y dx$, where C is the circle $x^2 + y^2 = 4$.

$$\text{Green's theorem is } \oint_C f(x,y) dx + g(x,y) dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

$$g(x,y) = x, \quad \frac{\partial g}{\partial x} = 1$$

$$f(x,y) = -y, \quad \frac{\partial f}{\partial y} = -1$$

$$\begin{aligned}
 \therefore \oint_C x dy - y dx &= \iint (1 - -1) dA \\
 &= 2 \iint dA \\
 &= 2 \cdot \text{Area of the circle} \\
 &= 2 \times \pi \times 2^2 = \underline{\underline{8\pi}}
 \end{aligned}$$

- 4) Determine whether the vector field $\vec{F} = 4(x^3 - x)\hat{i} + 4(y^3 - y)\hat{j} + 4(z^3 - z)\hat{k}$ is free of sources and sinks. If not locate them.

$$\begin{aligned}
 \text{div } \vec{F} &= \nabla \cdot \vec{F} = 4(3x^2 - 1) + 4(3y^2 - 1) + 4(3z^2 - 1) \\
 &= 12x^2 + 12y^2 + 12z^2 - 12
 \end{aligned}$$

$$\begin{aligned}
 \text{div } \vec{F} > 0 &\Rightarrow 12x^2 + 12y^2 + 12z^2 - 12 > 0 \\
 &\Rightarrow x^2 + y^2 + z^2 > 1
 \end{aligned}$$

$\therefore$ Exterior points of the unit sphere are the sources.

$\text{div } \vec{F} < 0 \Rightarrow x^2 + y^2 + z^2 < 1$, interior points of the unit sphere are the sinks.

$\text{div } \vec{F} = 0 \Rightarrow x^2 + y^2 + z^2 = 1$, points on the unit sphere are free of sources and sinks.

- 5) Show that the functions x , $x \ln x$ are linearly independent.

$$\begin{aligned}
 \text{Wronskian} = W &= \begin{vmatrix} x & x \ln x \\ 1 & x \cdot \frac{1}{x} + \ln x \end{vmatrix} = \begin{vmatrix} x & x \ln x \\ 1 & 1 + \ln x \end{vmatrix} \\
 &= x(1 + \ln x) - x \ln x \\
 &= x + x \ln x - x \ln x = x \neq 0
 \end{aligned}$$

$\therefore$ linearly independent.

6) Solve the differential equation $y'' + 4y' + 2.5y = 0$

Auxiliary equation is $m^2 + 4m + 2.5 = 0$

$$m = \frac{-4 \pm \sqrt{6}}{2}$$

$$y = c_1 e^{\frac{(-4+\sqrt{6})x}{2}} + c_2 e^{\frac{(-4-\sqrt{6})x}{2}}$$

7) Find the Laplace transform of $\sin 4t \cos 3t$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$L[\sin 4t \cos 3t] = L\left[\frac{1}{2} (\sin 7t + \sin t)\right]$$

$$= \frac{1}{2} [L(\sin 7t) + L(\sin t)]$$

$$= \frac{1}{2} \left[\frac{7}{s^2 + 49} + \frac{1}{s^2 + 1} \right]$$

8) Find the Laplace transform of $e^{-3t} u(t-1)$.

$$L[e^{-3t} u(t-1)] = L[e^{-3(t-1+1)} u(t-1)]$$

$$= L[e^{-3(t-1)-3} u(t-1)]$$

$$= L[e^{-3} \cdot e^{-3(t-1)} u(t-1)]$$

$$= e^{-3} L[e^{-3(t-1)} u(t-1)]$$

$$= e^{-3} \cdot \frac{e^{-s}}{s+3} = e^{-3-s} \cdot \frac{1}{s+3}$$

9) Determine the Fourier sine transform of $f(x) = 3x$, $0 < x < 6$.

$$F_s(s) = \sqrt{\frac{2}{\pi}} \int_0^6 f(x) \sin wx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^6 3x \sin wx \, dx$$

$$\begin{aligned}
 &= \sqrt{\frac{2}{\pi}} \cdot 3 \left[x \cdot \frac{-\cos \omega x}{\omega} - 1 \cdot \frac{\sin \omega x}{\omega^2} \right]_0^6 \\
 &= 3 \sqrt{\frac{2}{\pi}} \left[-\frac{6 \cos 6\omega}{\omega} + \frac{\sin 6\omega}{\omega^2} - 0 \right] \quad \left\{ \begin{array}{l} u = x, \quad v = \sin \omega x \\ u' = 1, \quad v_1 = -\frac{\cos \omega x}{\omega} \\ \quad \quad \quad v_2 = \frac{\sin \omega x}{\omega^2} \end{array} \right. \\
 &= 3 \sqrt{\frac{2}{\pi}} \left[\frac{\sin 6\omega}{\omega^2} - \frac{6 \cos 6\omega}{\omega} \right] \quad \int uv = uv_1 - u'v_2
 \end{aligned}$$

10) Find the Fourier sine integral of $f(x) = \begin{cases} \sin x, & 0 < x < \pi \\ 0, & x > \pi \end{cases}$

Fourier sine integral is

$$f(x) = \int_0^{\infty} B(\omega) \sin \omega x \, d\omega$$

$$B(\omega) = \frac{2}{\pi} \int_0^{\infty} f(x) \sin \omega x \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \sin x \sin \omega x \, dx \quad \left\{ \begin{array}{l} \sin A \sin B \\ = \frac{1}{2} [\cos(A-B) - \cos(A+B)] \end{array} \right.$$

$$= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} [\cos(x-\omega x) - \cos(x+\omega x)] \, dx$$

$$= \frac{1}{\pi} \left[\frac{\sin(x-\omega x)}{1-\omega} - \frac{\sin(x+\omega x)}{1+\omega} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\sin(\pi - \omega\pi)}{1-\omega} - \frac{\sin(\pi + \omega\pi)}{1+\omega} - 0 \right]$$

$$= \frac{1}{\pi} \left[\frac{\sin \omega\pi}{1-\omega} - \frac{\sin \omega\pi}{1+\omega} \right]$$

$$\left\{ \begin{array}{l} \sin(180 - \theta) = \sin \theta \\ \sin(180 + \theta) = -\sin \theta \end{array} \right.$$

$$= \frac{\sin \omega\pi}{\pi} \left[\frac{1}{1-\omega} + \frac{1}{1+\omega} \right]$$

$$= \frac{\sin \omega\pi}{\pi} \left(\frac{1+\omega + 1-\omega}{(1-\omega)(1+\omega)} \right) = \frac{\sin \omega\pi}{\pi} \times \frac{2}{1-\omega^2}$$

$$= \frac{2 \sin \omega\pi}{\pi(1-\omega^2)}$$

PART - B

MODULE - 1

- 11 a. Find the divergence and curl of the vector field

$$\vec{F}(x, y, z) = 3y\hat{i} + y^2x\hat{j} + yz^2\hat{k}$$

Sol: $\text{div } \vec{F} = \nabla \cdot \vec{F} = \frac{\partial}{\partial x}(3y) + \frac{\partial}{\partial y}(y^2x) + \frac{\partial}{\partial z}(yz^2)$

$$= 0 + 2yz + 2yz$$

$$\text{Curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3y & y^2x & yz^2 \end{vmatrix}$$

$$= \hat{i}[z^2 - 0] - \hat{j}[0 - y] + \hat{k}[y^2 - 3]$$

$$= z^2\hat{i} + y\hat{j} + (y^2 - 3)\hat{k}$$

- 11 b. Show that $\vec{F} = (\cos y + y \cos x)\hat{i} + (\sin x - x \sin y)\hat{j}$ is a conservative vector field. Hence find a potential function for it.

Sol: To show that $\vec{F}$ is a conservative field, it is enough we show, $\text{Curl } \vec{F} = 0$

$$\text{Curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \cos y + y \cos x & \sin x - x \sin y & 0 \end{vmatrix}$$

$$\text{Curl } \vec{F} = i(0-0) - j(0-0) + k[(\cos x - \sin y) - (-\sin y + \cos x)]$$

$$= 0i - 0j + 0k$$

$= 0 \Rightarrow \vec{F}$ is a conservative field.

Since $\vec{F}$ is a conservative field, $\vec{F} = \nabla \phi$

$$\text{i.e. } i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z} = (\cos y + y \cos x) i + (\sin x - x \sin y) j$$

$$\Rightarrow \frac{\partial \phi}{\partial x} = \cos y + y \cos x$$

Integrating w.r.t 'x' both sides: $\phi = x \cos y + y \sin x + \text{terms without } x$

$$\text{Also: } \frac{\partial \phi}{\partial y} = \sin x - x \sin y$$

Integrating w.r.t 'y' both sides: $\phi = y \sin x - x \cos y + \text{terms without } y$

$\therefore \phi = x \cos y + y \sin x + C$ which is the required potential function.

12
a.

Find the workdone by the force $\vec{F} = xy\hat{i} + yz\hat{j} + xz\hat{k}$ on a particle that moves along the curve $C: \vec{r}(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ where $0 \leq t \leq 1$

Sol:

$$\text{Workdone} = \int_C \vec{F} \cdot d\vec{r}$$

$$= \int_C (xy\hat{i} + yz\hat{j} + xz\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$= \int_C xy dx + yz dy + xz dz$$

$$= \int_C t \cdot t^2 dt + t^2 \cdot t^3 \cdot 2t dt + t \cdot t^3 \cdot 3t^2 dt$$

$$= \int_{t=0}^1 (t^3 + 2t^6 + 3t^6) dt$$

$$= \int_{t=0}^1 (t^3 + 5t^6) dt$$

$$= \left[\frac{t^4}{4} + 5 \frac{t^7}{7} \right]_0^1 = \frac{1}{4} + \frac{5}{7} = \underline{\underline{\frac{27}{28} \text{ units}}}$$

Given:

$$C: \vec{r}(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$$

$$\Rightarrow x=t, y=t^2, z=t^3$$

$$\Rightarrow dx=dt$$

$$dy=2t dt$$

$$dz=3t^2 dt$$

$$\Rightarrow t: 0 \rightarrow 1$$

12. b.

Show that $\int_C 3x^2 e^y dx + x^3 e^y dy$ is independent of the path and hence evaluate the integral from (0,0) to (3,2)

Sol:

Given line integral is independent of path if corresponding $\vec{F}$ is irrotational / conservative.

$$\text{Here: } \vec{F} = 3x^2 e^y \hat{i} + x^3 e^y \hat{j}$$

$$\nabla \times \bar{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2e^y & x^3e^y & 0 \end{vmatrix}$$

$$= \hat{i} [0 - 0] - \hat{j} [0 - 0] + \hat{k} [3x^2e^y - 3x^2e^y]$$

$$= 0$$

$\Rightarrow \bar{F}$ is conservative.

$\Rightarrow$ Given line integral is independent of the path.

To evaluate the line integral:

$$\int_{(0,0)}^{(3,2)} 3x^2e^y dx + x^3e^y dy = \phi(3,2) - \phi(0,0) \quad \text{--- ①}$$

To find ϕ :

Since $\nabla \times \bar{F} = 0$, we have $\bar{F} = \nabla \phi$

$$\text{i.e. } \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} = 3x^2e^y \hat{i} + x^3e^y \hat{j}$$

$$\Rightarrow \frac{\partial \phi}{\partial x} = 3x^2e^y \Rightarrow \int_{\text{w.r.t 'x'}}^{\text{ing}} : \phi = \frac{3x^3}{3} e^y + \text{terms without } x.$$

$$\Rightarrow \frac{\partial \phi}{\partial y} = x^3e^y \Rightarrow \int_{\text{w.r.t 'y'}}^{\text{ing}} : \phi = x^3e^y + \text{terms without } y.$$

$$\therefore \phi = x^3e^y + C$$

$$\therefore \phi(3,2) = 3^3 e^2 + C = 27e^2 + C$$

$$\phi(0,0) = 0 + C = C$$

Substituting these values in ① ;

$$\int_{(0,0)}^{(3,2)} 3x^2 e^y dx + x^3 e^y dy = 27e^2 + C - C = \underline{\underline{27e^2}}$$

MODULE - 2

13a. Using Green's theorem, evaluate the line integral $\int_C (xy + y^2) dx + x^2 dy$ where C is bounded by $y = x$ and $y = x^2$ and positively oriented.

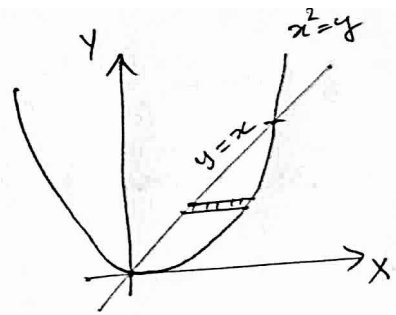
Sol: Green's theorem: $\int_C f dx + g dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy$

Comparing given line integral with LHS of Green's theorem, we have: $f = xy + y^2$, $g = x^2$

$$\therefore \frac{\partial f}{\partial y} = x + 2y, \quad \frac{\partial g}{\partial x} = 2x$$

$$\begin{aligned} \therefore \int_C (xy + y^2) dx + x^2 dy &= \iint_R 2x - (x + 2y) dx dy \\ &= \iint_R (x - 2y) dx dy. \end{aligned}$$

$$= \int_{y=0}^1 \int_{x=y}^{\sqrt{y}} (x-2y) dx dy$$



$$= \int_{y=0}^1 \left[\frac{x^2}{2} - 2yx \right]_{x=y}^{\sqrt{y}} dy$$

$$= \int_{y=0}^1 \left(\frac{y}{2} - 2y \cdot \sqrt{y} \right) - \left(\frac{y^2}{2} - 2y \cdot y \right) dy$$

$$= \int_{y=0}^1 \left(\frac{y}{2} - 2y^{\frac{3}{2}} - \frac{y^2}{2} + 2y^2 \right) dy$$

$$= \left[\frac{y^2}{4} - 2 \frac{y^{\frac{5}{2}}}{\frac{5}{2}} - \frac{y^3}{6} + 2 \frac{y^3}{3} \right]_{y=0}^1$$

$$= \frac{1}{4} - \frac{4}{5} - \frac{1}{6} + \frac{2}{3}$$

$$= \underline{\underline{-\frac{1}{20}}}$$

[OR]

$$\int_C (xy + y^2) dx + x^2 dy = \int_{x=0}^1 \int_{y=x^2}^x (x-2y) dy dx = \underline{\underline{-\frac{1}{20}}}$$

Meeting point

$$y=x, y=x^2$$

$$\Rightarrow x=x^2$$

$$\Rightarrow x^2-x=0$$

$$\Rightarrow x(x-1)=0$$

$$\Rightarrow x=0, x=1$$

$$\downarrow$$

$$y=0$$

$$y=1$$

$$(0,0) + (1,1)$$

13. Evaluate the surface integral $\iint_{\sigma} z^2 ds$, where σ is the portion of the cone $z = \sqrt{x^2 + y^2}$ between the planes $z=1$ and $z=3$

Sol: Projecting the surface to xy -plane,

$$\iint_{\sigma} f(x, y, z) ds = \iint_R f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA.$$

$$\iint_{\sigma} z^2 ds = \iint_R (x^2 + y^2) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dx dy. \quad \text{--- ①}$$

$$z = \sqrt{x^2 + y^2} \quad \therefore \frac{\partial z}{\partial x} = \frac{1}{2\sqrt{x^2 + y^2}} \cdot 2x = \frac{x}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial z}{\partial y} = \frac{1}{2\sqrt{x^2 + y^2}} \cdot 2y = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\begin{aligned} \therefore \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} &= \sqrt{\frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2} + 1} \\ &= \sqrt{\frac{x^2 + y^2}{x^2 + y^2} + 1} = \sqrt{2} \end{aligned}$$

$\therefore$ equ. ① becomes,

$$\iint_{\sigma} z^2 ds = \iint_R (x^2 + y^2) \sqrt{2} dx dy.$$

Converting to polar coordinates,

$$\int_0^{\sigma} \int_0^{2\pi} \int_1^3 r^2 \sqrt{2} \, r \, dr \, d\theta$$

$$= \int_{\theta=0}^{2\pi} \int_{r=1}^3 r^3 \sqrt{2} \, dr \, d\theta$$

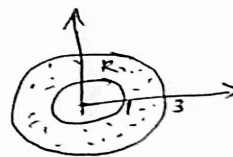
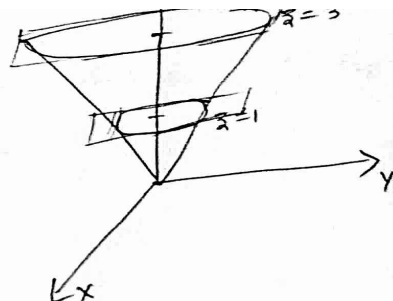
$$= \int_{\theta=0}^{2\pi} \sqrt{2} \left[\frac{r^4}{4} \right]_{r=1}^3 d\theta$$

$$= \int_{\theta=0}^{2\pi} \frac{\sqrt{2}}{4} [3^4 - 1^4] d\theta$$

$$= \frac{\sqrt{2}}{4} \times 80 \left[\theta \right]_0^{2\pi}$$

$$= \sqrt{2} \times 20 [2\pi - 0]$$

$$= \underline{\underline{40\pi \cdot \sqrt{2}}}$$



14

- a. Evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$, where S is the surface of the cylinder $x^2 + y^2 = 4$, $z = 0$, $z = 3$ where,
 $\vec{F} = (2x - y)\hat{i} + (2y - z)\hat{j} + z^2\hat{k}$

Sol: We know, by divergence theorem,

$$\iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dv \quad \text{--- ①}$$

$$\begin{aligned} \nabla \cdot \vec{F} &= \frac{\partial}{\partial x}(2x - y) + \frac{\partial}{\partial y}(2y - z) + \frac{\partial}{\partial z}(z^2) \\ &= 2 + 2 + 2z = 4 + 2z \end{aligned}$$

① becomes,

$$\iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V (4 + 2z) \, dv$$

Converting to cylindrical coordinates

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^2 \int_{z=0}^3 (4 + 2z) r \, dz \, dr \, d\theta$$

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^2 \left[4rz + 2r \frac{z^2}{2} \right]_{z=0}^3 dr \, d\theta$$

$$\left\{ \begin{array}{l} \text{Cylindrical} \\ \text{Coordinates,} \\ x = r \cos \theta \\ y = r \sin \theta \\ z = z \\ dv = r \, dz \, dr \, d\theta \end{array} \right.$$

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^2 (12r + 9r) dr d\theta$$

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^2 21r dr d\theta$$

$$= \int_{\theta=0}^{2\pi} \left[21 \frac{r^2}{2} \right]_{r=0}^2 d\theta$$

$$= \int_{\theta=0}^{2\pi} \frac{21}{2} \cdot (4-0) d\theta = 42 \left[\theta \right]_0^{2\pi} = \underline{\underline{84\pi}}$$

14
b.

Apply Stokes' theorem to evaluate

$$\int_C (x-y) dx + (y-z) dy + (z-x) dz \text{ where } C \text{ is the}$$

boundary of the portion of the plane $x+y+z=1$ in the 1st octant.

SA: Stokes theorem:

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dA.$$

Comparing given line integral, with LHS of Stokes theorem,

we've

$$\vec{F} = (x-y)\hat{i} + (y-z)\hat{j} + (z-x)\hat{k}$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x-y & y-z & z-x \end{vmatrix}$$

$$= \hat{i}[0-1] - \hat{j}[-1-0] + \hat{k}[0-1]$$

$$= -\hat{i} + \hat{j} - \hat{k}$$

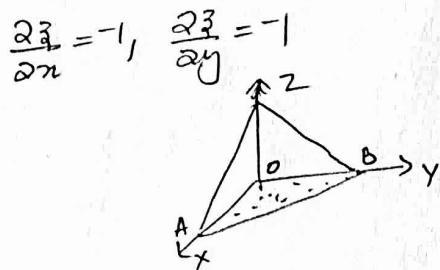
Using Stokes theorem,

$$\int_C (x-y)dx + (y-z)dy + (z-x)dz = \iint_{\sigma} (-\hat{i} + \hat{j} - \hat{k}) \cdot \hat{n} dA \quad \text{--- (1)}$$

$$\hat{n} = -\frac{\partial z}{\partial x}\hat{i} - \frac{\partial z}{\partial y}\hat{j} + \hat{k} \quad \text{where } z = 1-x-y \quad \because x+y+z=1$$

$$= -(-1)\hat{i} - (-1)\hat{j} + \hat{k}$$

$$= \hat{i} + \hat{j} + \hat{k}$$



$\therefore$ (1) becomes,

$$\int_C (x-y)dx + (y-z)dy + (z-x)dz = \iint_{\sigma} (\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{i} + \hat{j} + \hat{k}) dA$$
$$= \iint_{\sigma} 3 dA$$

$$= 3 \times \text{Area of } \triangle OAB$$

$$= 3 \times \frac{1}{2}bh = 3 \times \frac{1}{2} \times 1 \times 1$$

$$= \frac{3}{2} //$$

module - 3

15)
a)

Solve the Cauchy - Euler differential equ
 $(x^2 D^2 - 3x D + 10)y = 0$

Ans

$$\begin{aligned} AE \Rightarrow m^2 + (a-1)m + b &= 0 & \left| \begin{array}{l} a = -3 \\ b = 10 \end{array} \right. \\ \Rightarrow m^2 + (-3-1)m + 10 &= 0 \\ \Rightarrow m^2 - 4m + 10 &= 0 \end{aligned}$$

$$\Rightarrow m = 2 \pm i\sqrt{6} \quad (\text{Roots complex})$$

$$\alpha = 2, \quad \beta = \sqrt{6}$$

$$\therefore y = x^\alpha [c_1 \cos(\beta \log x) + c_2 \sin(\beta \log x)]$$

$$y = x^2 [c_1 \cos \sqrt{6} \log x + c_2 \sin \sqrt{6} \log x]$$

b)

Solve the initial value problem $y'' - 2y' + 5y = 0$
 $y(0) = -3, y'(0) = 1$

Ans:

$$y'' - 2y' + 5y = 0 \quad \text{--- (1)}$$

$$AE \Rightarrow m^2 - 2m + 5 = 0$$

$$\Rightarrow m = 1 \pm 2i \quad (\text{complex})$$

$$\alpha = 1, \quad \beta = 2$$

$$\therefore y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

$$y = e^x (c_1 \cos 2x + c_2 \sin 2x) \quad \text{--- (2)}$$

$$\text{apply } y(0) = -3 \text{ in (2)} \Rightarrow -3 = e^0 (c_1) \Rightarrow \underline{\underline{c_1 = -3}}$$

$$(2) \Rightarrow y' = e^x (c_1 x - 2 \sin 2x + c_2 x 2 \cos 2x) + (c_1 \cos 2x + c_2 \sin 2x) \times e^x. \text{ --- (3)}$$

apply $y'(0) = 1$ in (3)

$$(3) \Rightarrow 1 = (0 + 2c_2) + (c_1)$$

$$\Rightarrow 1 = 2c_2 + c_1$$

$$\Rightarrow 1 = 2c_2 - 3$$

$$\Rightarrow 2c_2 = 4 \Rightarrow \underline{\underline{c_2 = 2}}$$

Substitute c_1, c_2 in (2).

$$\therefore (2) \Rightarrow y = e^x (-3 \cos 2x + 2 \sin 2x)$$

16)
a)

By the method of undetermined coefficient
solve $y'' + y' - 2y = \sin x$.

Ans:

$$y'' + y' - 2y = \sin x \text{ --- (1)}$$

$$\text{soln of (1) is } y = y_h + y_p \text{ --- (2)}$$

To find y_h

$$\text{L.H.S of (1) } = 0$$

$$y'' + y' - 2y = 0$$

$$\text{A.E.} \Rightarrow m^2 + m - 2 = 0$$

$$\Rightarrow m = 1, -2. \text{ (Real distinct)}$$

$$\therefore \underline{y_h = c_1 e^x + c_2 e^{-2x}} \quad \text{--- (3)}$$

To find y_p

R.H.S of (1) is " $\sin x$."

choose ~~$y_p = \sin x$~~ ~~$y_p = L \cos x$~~

choose $y_p = L \cos \omega x + M \sin \omega x$ R.H.S is $\sin x$.
 $\omega = 1$

$$y_p = L \cos x + M \sin x \quad \text{--- (4)}$$

$$y_p' = -L \sin x + M \omega \cos x$$

$$y_p'' = -L \omega \sin x - M \sin x$$

$$\therefore \textcircled{1} \Rightarrow y_p'' + y_p' - 2y_p = \sin x$$

$$\therefore \Rightarrow -L \cos x - m \sin x - L \sin x + m \cos x - 2L \cos x - 2m \sin x = \sin x.$$

$$\Rightarrow -3L \cos x - 3m \sin x - L \sin x + m \cos x = \sin x.$$

$$\Rightarrow \begin{array}{ll} \text{equating 'sin x'} & \text{equating 'cos x'} \\ -3m - L = 1 & -3L + m = 0 \end{array}$$

$$\text{solve: } \begin{array}{r} -3m - L = 1 \\ + m - 3L = 0 \end{array}$$

$$\therefore \underline{\underline{L = -\frac{1}{10}}}, \underline{\underline{m = -\frac{3}{10}}}$$

$$\therefore \textcircled{4} \Rightarrow y_p = \underline{\underline{-\frac{1}{10} \cos x - \frac{3}{10} \sin x}}. \quad \text{--- } \textcircled{5}$$

$\therefore$ substitute $\textcircled{3}, \textcircled{5}$ in $\textcircled{2}$

$$\therefore \underline{\underline{y = c_1 e^x + c_2 e^{-2x} - \frac{1}{10} \cos x - \frac{3}{10} \sin x}}$$

b) Using method of variation of parameters
solve $\frac{d^2y}{dx^2} + 4y = \sec 2x$.

Ans:

$$y'' + 4y = \sec 2x \quad \text{--- (1)}$$

$$\text{soln of (1) is } y = y_h + y_p \quad \text{--- (2)}$$

To find y_h

$$y'' + 4y = 0$$

$$AE \Rightarrow m^2 + 4 = 0$$

$$\Rightarrow m = \pm 2i \text{ (complex)}$$

$$\underline{y_h = (c_1 \cos 2x + c_2 \sin 2x)} \quad \text{--- (3)}$$

To find y_p (using method of variation of parameters)

$$y_p = -y_1 \int \frac{y_2 \cdot r}{w} \cdot dx + y_2 \int \frac{y_1 \cdot r}{w} dx \quad \text{--- (4)}$$

$$\text{from (3)} \Rightarrow \underline{y_1 = \cos 2x}, \quad \underline{y_2 = \sin 2x}$$

$$w = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix} = 2[\cos^2 2x + \sin^2 2x] = \underline{2}$$

$$\text{from (1)} \Rightarrow \underline{r = \sec 2x}$$

④ $\Rightarrow$

$$y_p = -\cos 2x \int \frac{\sin 2x \cdot \sec 2x}{2} dx + \sin 2x \int \frac{\cos 2x \cdot \sec 2x}{2} dx$$

$$= -\frac{\cos 2x}{2} \int \frac{\sin 2x}{\cos 2x} dx + \frac{\sin 2x}{2} \int \frac{\cos 2x}{\cos 2x} dx$$

$$= -\frac{\cos 2x}{2} \int \frac{\sin 2x}{\cos 2x} dx + \frac{\sin 2x}{2} \int dx$$

$$= -\frac{\cos 2x}{2} \times \frac{1}{2} \log(\cos 2x) + \frac{\sin 2x}{2} (x)$$

$$= \frac{1}{4} \cos 2x \cdot \log(\cos 2x) + \frac{x \sin 2x}{2}$$

Module IV

(17)

(a) Find the Laplace transform of $\cos^2 t$.

Soln.

$$\begin{aligned} L(\cos^2 t) &= L\left(\frac{1 + \cos 2t}{2}\right) \\ &= \frac{1}{2} [L(1) + L(\cos 2t)] \\ &= \frac{1}{2} \left[\frac{1}{s} + \frac{s}{s^2 + 4} \right] \end{aligned}$$

(b) Find the inverse Laplace transform of $\frac{3s+2}{(s-1)(s^2+1)}$

Soln.

$$\begin{aligned} \frac{3s+2}{(s-1)(s^2+1)} &= \frac{A}{s-1} + \frac{Bs+C}{s^2+1} \\ &= \frac{A(s^2+1) + (Bs+C)(s-1)}{(s-1)(s^2+1)} \\ \Rightarrow 3s+2 &= A(s^2+1) + (Bs+C)(s-1) \quad \text{--- (1)} \end{aligned}$$

$$s=1 \Rightarrow 5 = 2A \Rightarrow A = 5/2.$$

Equating ' s^2 ' coefficients both sides,

$$0 = A + B \Rightarrow B = -5/2$$

Equating constant terms both sides,

$$2 = A - C$$

$$2 = \frac{5}{2} - C \Rightarrow C = \frac{5}{2} - 2 = 1/2$$

$$\therefore \frac{3s+2}{(s-1)(s^2+1)} = \frac{5/2}{s-1} + \frac{(-5/2)s}{s^2+1} + \frac{1/2}{s^2+1}$$

$$\begin{aligned} \mathcal{L}^{-1} \left[\frac{3s+2}{(s-1)(s^2+1)} \right] &= \frac{5}{2} \mathcal{L}^{-1} \left[\frac{1}{s-1} \right] - \frac{5}{2} \mathcal{L}^{-1} \left[\frac{s}{s^2+1} \right] \\ &\quad + \frac{1}{2} \mathcal{L}^{-1} \left[\frac{1}{s^2+1} \right] \end{aligned}$$

$$= \frac{5}{2} e^t - \frac{5}{2} \cos t + \frac{1}{2} \sin t$$

(18)

(a) Using Laplace transform solve

$$y'' + 5y' + 6y = e^{-2t} \text{ given that}$$

$$y(0) = y'(0) = 1.$$

Soln: $L(y'' + 5y' + 6y) = L(e^{-2t})$

$$L(y'') + 5L(y') + 6L(y) = L(e^{-2t})$$

$$[s^2 L(y) - s y(0) - y'(0)] + 5[s L(y) - y(0)] + 6L(y) = \frac{1}{s+2}$$

$$(s^2 L(y) - s - 1) + 5(s L(y) - 1) + 6L(y) = \frac{1}{s+2}$$

$$L(y) [s^2 + 5s + 6] - s - 1 - 5 = \frac{1}{s+2}$$

$$L(y) [s^2 + 5s + 6] = \frac{1}{s+2} + s + 6$$

$$L(y) = \frac{1}{(s+2)(s^2 + 5s + 6)} + \frac{s+6}{s^2 + 5s + 6}$$

$$= \frac{1}{(s+2)(s+2)(s+3)} + \frac{s+6}{(s+2)(s+3)}$$

$$L(y) = \frac{(s+2)(s+6) + 1}{(s+2)^2(s+3)}$$

$$L(y) = \frac{A}{s+2} + \frac{B}{(s+2)^2} + \frac{C}{s+3}$$

$$\Rightarrow \frac{(s+2)(s+6) + 1}{(s+2)^2(s+3)} = \frac{A(s+2)(s+3) + B(s+3) + C(s+2)^2}{(s+2)^2(s+3)}$$

$$(s+2)(s+6) + 1 = A(s+2)(s+3) + B(s+3) + C(s+2)^2 \quad \text{--- (1)}$$

$$\text{Let } s = -2 \Rightarrow \boxed{1 = B}$$

$$s = -3 \Rightarrow (-1)(3) + 1 = C \Rightarrow \boxed{C = -2}$$

Equating coefficients of s^2 ,

$$1 = A + C \Rightarrow \boxed{A = 1 + 2 = 3}$$

$$\therefore L(y) = \frac{3}{s+2} + \frac{1}{(s+2)^2} + \frac{-2}{s+3}$$

$$y = L^{-1}\left[\frac{3}{s+2}\right] + L^{-1}\left[\frac{1}{(s+2)^2}\right] + L^{-1}\left[\frac{-2}{s+3}\right]$$

$$y = \underline{\underline{3 \cdot e^{-2t} + t e^{-2t} - 2 e^{-3t}}}$$

(b) Find the inverse Laplace transform of $\frac{s}{(s^2+a^2)^2}$ using Convolution.

Soln

$$\frac{s}{(s^2+a^2)^2} = \frac{s}{s^2+a^2} \cdot \frac{1}{s^2+a^2}$$

$$\text{Let } F(s) = \frac{s}{s^2+a^2} \Rightarrow f(t) = \cos at$$

$$G(s) = \frac{1}{s^2+a^2} \Rightarrow g(t) = \frac{1}{a} \sin at$$

By Convolution theorem,

$$\begin{aligned} L^{-1}[F(s) \cdot G(s)] &= f(t) * g(t) \\ &= \int_0^t f(u) g(t-u) du \\ &= \int_0^t \cos au \cdot \frac{1}{a} \sin a(t-u) du \\ &= \frac{1}{a} \int_0^t \cos au \cdot \sin(at-au) du \\ &= \frac{1}{a} \int_0^t \frac{\sin at + \sin(at-2au)}{2} du \\ &= \frac{1}{2a} \left[u \sin at - \frac{\cos(at-2au)}{-2a} \right]_0^t \end{aligned}$$

$$= \frac{1}{2a} \left[\left(t \sin at + \frac{\cos(at - 2at)}{2a} \right) - \left(0 + \frac{\cos at}{2a} \right) \right]$$

$$= \frac{1}{2a} \left[t \sin at + \frac{\cos(-at)}{2a} - \frac{\cos at}{2a} \right]$$

$$= \frac{1}{2a} \left[t \sin at + \frac{\cos at}{2a} - \frac{\cos at}{2a} \right]$$

$$= \frac{1}{2a} (t \sin at)$$

Hence $\underline{\underline{L^{-1} \left[\frac{s}{(s^2 + a^2)^2} \right] = \frac{t \sin at}{2a}}}$

Module V

(19)

(a) Find the Fourier Transform and integral representation of

$$f(x) = \begin{cases} 1, & |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$

Hence show that $\int_0^{\infty} \frac{\sin \omega}{\omega} d\omega = \pi/2$.

Fourier Transform

Soln -

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 1 \cdot e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (\cos \omega x dx - i \sin \omega x) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 \cos \omega x dx - \frac{1}{\sqrt{2\pi}} i \int_{-1}^1 \sin \omega x dx$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^1 \cos \omega x dx - 0$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{\sin \omega x}{\omega} \right]_0^1 = \underline{\underline{\sqrt{\frac{2}{\pi}} \frac{\sin \omega}{\omega}}}$$

Fourier Integral Representation

$$f(x) = \int_0^{\infty} (A(\omega) \cos \omega x + B(\omega) \sin \omega x) d\omega$$

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \omega x dx$$

$$= \frac{1}{\pi} \int_{-1}^1 1 \cdot \cos \omega x dx = \frac{2}{\pi} \int_0^1 \cos \omega x dx$$

$$= \frac{2}{\pi} \left[\frac{\sin \omega x}{\omega} \right]_0^1 = \frac{2}{\pi} \frac{\sin \omega}{\omega}$$

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \omega x dx$$

$$= \frac{1}{\pi} \int_{-1}^1 1 \cdot \sin \omega x dx = \underline{\underline{0}}$$

$$\therefore f(x) = \int_0^{\infty} \frac{2}{\pi} \frac{\sin \omega}{\omega} \cos \omega x d\omega \quad \text{--- (1)}$$

let $x=0$ in (1),

$$1 = \int_0^{\infty} \frac{2}{\pi} \frac{\sin \omega}{\omega} d\omega$$

$$\Rightarrow \underline{\underline{\int_0^{\infty} \frac{\sin \omega}{\omega} d\omega = \pi/2}}$$

(b) Find the Fourier sine transform and inverse transform of e^{-ax} , $x > 0$, $a > 0$

Soln Fourier sine transform,

$$\begin{aligned} f_s^{\wedge}(\omega) &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin \omega x \, dx \\ &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-ax} \sin \omega x \, dx \\ &= \sqrt{\frac{2}{\pi}} \left(\frac{\omega}{a^2 + \omega^2} \right) \end{aligned}$$

Inverse Fourier Sine Transform,

$$\begin{aligned} f(x) &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f_s^{\wedge}(\omega) \sin \omega x \, d\omega \\ &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \sqrt{\frac{2}{\pi}} \left(\frac{\omega}{a^2 + \omega^2} \right) \sin \omega x \, d\omega \\ &= \frac{2}{\pi} \int_0^{\infty} \frac{\omega}{a^2 + \omega^2} \sin \omega x \, d\omega \end{aligned}$$

(20)

(a) Find the complex Fourier transform

$$\text{of } f(x) = \begin{cases} \sin x & , |x| \leq a, a > 0 \\ 0 & - |x| > a \end{cases}$$

Soln Fourier transform,

$$f^{\wedge}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-a}^a \sin x e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-a}^a \sin x (\cos \omega x - i \sin \omega x) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-a}^a \sin x \cos \omega x dx -$$

$$\frac{i}{\sqrt{2\pi}} \int_{-a}^a \sin x \sin \omega x dx$$

$$= -\frac{2i}{\sqrt{2\pi}} \int_0^a \sin x \sin \omega x dx$$

$$= \frac{-i}{\sqrt{2\pi}} \int_0^a [\cos (1-\omega)x - \cos (1+\omega)x] dx$$

$$= \frac{-j}{\sqrt{2\pi}} \left[\frac{\sin(1-\omega)x}{1-\omega} - \frac{\sin(1+\omega)x}{1+\omega} \right]_0^a$$

$$= \frac{-j}{\sqrt{2\pi}} \left[\left(\frac{\sin(1-\omega)a}{1-\omega} - \frac{\sin(1+\omega)a}{1+\omega} \right) - 0 \right]$$

$$= \frac{j}{\sqrt{2\pi}} \left[\frac{\sin(1+\omega)a}{1+\omega} - \frac{\sin(1-\omega)a}{1-\omega} \right]$$

(b) Find the Fourier transform and integral representation of

$$f(x) = \begin{cases} 1-x^2, & |x| \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Soln

Fourier transform,

$$f^{\wedge}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1-x^2) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1-x^2) (\cos \omega x - i \sin \omega x) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1-x^2) \cos \omega x dx -$$

$$\frac{i}{\sqrt{2\pi}} \int_{-1}^1 (1-x^2) \sin \omega x dx$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^1 (1-x^2) \cos \omega x dx - 0$$

$$= \sqrt{\frac{2}{\pi}} \left[(1-x^2) \left(\frac{\sin \omega x}{\omega} \right) - (-2x) \cdot \left(-\frac{\cos \omega x}{\omega^2} \right) + (-2) \left(-\frac{\sin \omega x}{\omega^3} \right) \right]_0^1$$

$$= \sqrt{\frac{2}{\pi}} \left[\left(0 - 2 \frac{\cos \omega}{\omega^2} + 2 \frac{\sin \omega}{\omega^3} \right) - 0 \right]$$

$$= 2\sqrt{\frac{2}{\pi}} \left[\frac{\sin \omega}{\omega^3} - \frac{\cos \omega}{\omega^2} \right]$$

$$= 2\sqrt{\frac{2}{\pi}} \left[\frac{\sin \omega - \omega \cos \omega}{\omega^3} \right]$$

Fourier Integral representation,

$$f(x) = \int_0^{\infty} (A(\omega) \cos \omega x + B(\omega) \sin \omega x) d\omega$$

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \omega x dx$$

$$= \frac{1}{\pi} \int_{-1}^1 (1-x^2) \cos \omega x dx$$

$$= \frac{2}{\pi} \int_0^1 (1-x^2) \cos \omega x dx$$

$$= \frac{2}{\pi} \left[(1-x^2) \left(\frac{\sin \omega x}{\omega} \right) - (-2x) \left(\frac{\cos \omega x}{\omega^2} \right) + (-2) \left(-\frac{\sin \omega x}{\omega^3} \right) \right]_0^1$$

$$= \frac{2}{\pi} \left[\left(-2 \frac{\cos \omega}{\omega^2} + 2 \frac{\sin \omega}{\omega^3} \right) - 0 \right]$$

$$= \frac{4}{\pi} \left[\frac{\sin \omega}{\omega^3} - \frac{\cos \omega}{\omega^2} \right]$$

$$= \frac{4}{\pi} \left[\frac{\sin \omega - \omega \cos \omega}{\omega^3} \right]$$

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \omega x dx$$

$$= \frac{1}{\pi} \int_{-1}^1 (1-x^2) \sin \omega x dx$$

$$= \underline{\underline{0}}$$

$$\therefore f(x) = \int_0^{\infty} \frac{4}{\pi} \left(\frac{\sin \omega - \omega \cos \omega}{\omega^3} \right) \cos \omega x d\omega$$

$$= \underline{\underline{\frac{4}{\pi} \int_0^{\infty} \left(\frac{\sin \omega - \omega \cos \omega}{\omega^3} \right) \cos \omega x d\omega}}$$